

Freight Rate Prediction - Approach Report

Spotter Machine Learning Engineer assessment

1. Task and data

Predict **posted_rate** (USD) for the 12,000 loads in validation.csv and a 31-day December forecast for one fixed lane (Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lb). The labelled data (train-test.csv) has 48,000 loads.

- **The split is temporal.** Training covers 2025-01-01 to 2025-10-31; validation covers 2025-11-01 to 2025-12-31. The real task is a forecast into unseen months.
- Rate depends mainly on distance (correlation 0.91), and **rate per mile falls as distance grows**: median \$2.87/mile under 150 miles down to \$1.91/mile above 2,000 miles (normal loads). The model therefore predicts rate per mile and multiplies by distance.
- Validation contains 8 cities never seen in training (Allentown, Charlotte, Chicago, Jackson, Knoxville, Laredo, Norfolk, San Diego); 12.2% of validation loads are on lanes never seen in training. Equipment mix is nearly identical in both files.

2. Validation and split approach

A random split would place days from the same weeks in both train and test, and would not reflect the task, which is to predict months that come after all the training data. All validation is therefore time-aware:

- **Forward-looking folds (3):** train on all earlier months and predict the next two - May-Jun, Jul-Aug, Sep-Oct. This mirrors the real setup (train Jan-Oct, predict Nov-Dec).
- **Leave-one-month-out (10):** hold out each month and train on the other nine. It is not forward-looking, but it covers 10 situations instead of 3 and shows which months break which model.
- Everything learned from labels (typical rate per mile, the corrupted-label rule) is recomputed from the training portion of each split only.
- Day-level market features are built from feature columns only (never from posted_rate), so they can also be computed for validation days and December.
- **Metric:** MAE in dollars on holdout rows with uncorrupted labels. The hidden scoring metric is not stated. A single holdout was found to be misleading: the winning model changes from fold to fold, which is why several folds are used.

3. Data-quality issues and how they were handled

- **Negative weights:** 292 training rows. Their absolute values have the same range and median as positive weights (5,000-47,500; median about 31,800 vs 31,500), so they are treated as sign errors and made positive.
- **Missing values:** weight 300 rows (0.6%), market_index 374 rows (0.8%), spread evenly over months (no pattern). Weight is left missing for LightGBM, which handles it natively. market_index is essentially one value per day (within-day std 0.025 vs 0.167 across days), so the day-level median is used.
- **Clipping:** 1,191 weights sit exactly at 47,500 and 48 distances exactly at 70 - values look capped or floored. Left unchanged.
- **Corrupted rates:** about 1.4% of rows (677 of 48,000) have a rate per mile far from normal (below \$1 or above \$5, from \$0.33 to \$14.13 per mile). They are spread evenly across equipment and months, so they behave like random label noise. A load is flagged when its rate per mile is below 0.5x or above 2.0x the median for its

equipment and distance band; in January-August the ratio distribution has an empty gap between about 0.34 and 0.86 and between 1.19 and 2.95, so the cutoffs are not sensitive. Flagged rows are excluded from **training only** and from the holdout used to compute the reported MAE.

4. Model

LightGBM (gradient-boosted trees) predicts **log(rate per mile)** with an L1 loss, which limits the influence of leftover noise; the dollar prediction is exp(prediction) times distance. Four feature variants are trained and their log predictions are averaged:

- **A:** distance, equipment, weight.
- **B:** A + the load's own market_index and quote_signal.
- **C:** A + **mi_day**, the median market_index of that day.
- **D:** A + mi_day + quote_signal.

Findings that shaped the model. The overall rate level drifts month to month (median rate per mile rose from \$2.03 in January to \$2.25 in June), so models that cannot see the market fail when the level moves (model A in May-Jun). market_index tracks the rate level well (the rate level is -2.4% in the lowest fifth of days and +3.5% in the highest; day-level correlation 0.70), but the link is **not stable over time**: January and September have similar values (0.93 and 0.89) while the rate level is -5.4% in January and +0.6% in September, which makes model C fail in Sep-Oct. quote_signal has a U-shaped relationship with rates (+1.9% at the lowest values, between -1.5% and -0.4% in the middle, +3.1% at the highest), so its plain correlation (0.07) understates it.

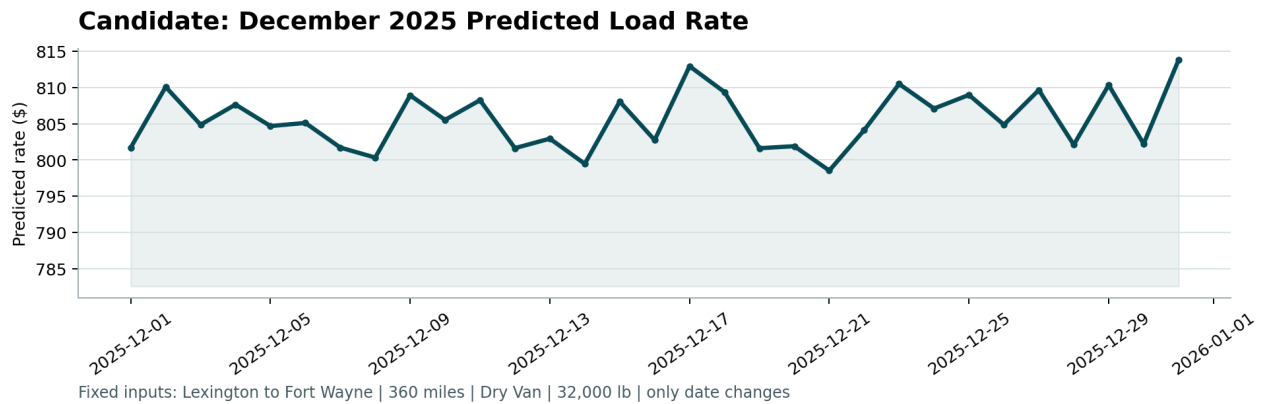
5. Results (MAE in dollars, clean-label holdout rows)

Model	Fwd mean	Fwd worst fold	LOMO mean	LOMO worst month
Baseline (median rate/mile by group)	107.7	145.7	-	-
A: distance, equipment, weight	91.2	147.1	83.8	140.4
B: A + market_index + quote_signal	69.2	87.9	57.9	117.1
C: A + mi_day	87.1	118.2	80.6	115.2
D: A + mi_day + quote_signal	71.1	91.2	59.9	115.0
Average of A-D (submitted)	68.9	72.3	63.1	93.1

Fwd = forward-looking folds; LOMO = leave-one-month-out. The baseline was only evaluated on the forward folds.

Why the average of four was submitted. Model B has the best mean MAE, but it (and D) does worst on August (MAE 117 and 115 vs 60 and 57 for A and C), and August is the only training month whose quote_signal profile resembles November and December (mean / std of quote_signal: August 2.052 / 0.223, November 2.055 / 0.220, December 2.048 / 0.219; every other month has a std of 0.25-0.29). Averaging models whose mistakes differ costs about \$5 of MAE in a typical month but cuts the worst-case error (forward worst fold 72.3 vs 87.9-147.1 for the single models). The blend was chosen after seeing these results, so its numbers are slightly optimistic; the claim is that it is the safer choice, not a proven best.

6. December forecast chart (produced by the provided score.py)



December has no training data and the chart input contains only the date, so the model uses the market values of the validation loads on the same date. Predictions range from \$798.53 to \$813.78 (mean \$805.50, about \$2.24 per mile) - a nearly flat line (the y-axis is zoomed; the swing is about 2%). The model has no December-specific information, so the small day-to-day variation is weak evidence and the level is the more reliable part.

7. Limitations

- The November-December rate level is extrapolated. Because the link between market_index and rates drifted (January vs September differ by about 6 points of rate level, roughly \$140 on a typical load), this is the largest source of uncertainty.
- The hidden evaluation metric is unknown; if the hidden labels contain the same 1.4% corruption, every submission's error will be dominated by it.
- Unseen validation cities are handled only through distance, equipment and market features; city identifiers and coordinates were not used (coordinates do not match real city locations).
- The mechanism behind quote_signal's effect was identified empirically but is not understood.